

Proteins

Definition: Proteins are organic molecules that are made up of the elements oxygen, hydrogen, carbon and nitrogen. Sulfur MAY also be present.

#BiologyDefinitions

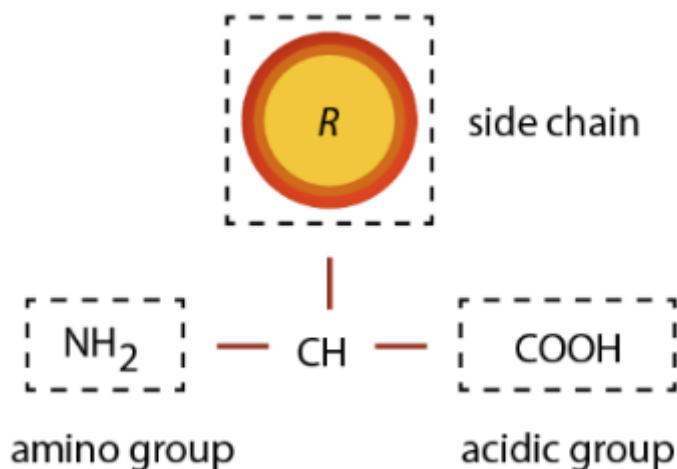
- Always present in all cells
- Molecules are the largest and most complicated of all food substances
- commonly found in meat of animals

*Amino Acids - Building Blocks of Proteins

- protein molecule $\Rightarrow$ built up from simpler molecules called amino acids

Amino acid is made up of:

- amino group (NH_2)
- side chain (denoted by R, which may sometimes contain sulfur)
- acidic group ($COOH$)



20 naturally occurring amino acids

$\rightarrow$ differ because of different R groups

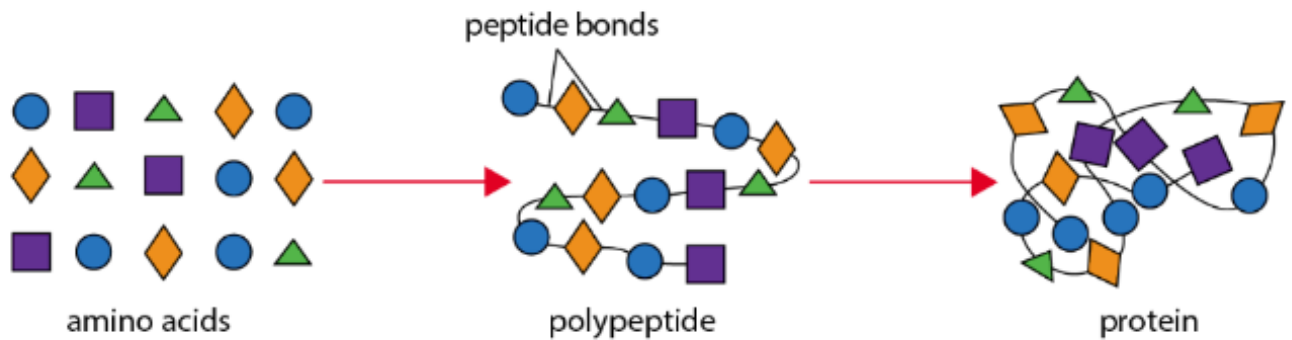
$\rightarrow$ Amino acids combined in various ways to form millions of diff proteins molecules

Long Chains of Amino Acids fold to give proteins a 3D Shape

$\rightarrow$ many amino acids $\rightarrow$ joined in linear manner to form polypeptide

$\rightarrow$ polypeptides may be linked to form even longer chains of amino acids

→ Protein molecule → made up of 1 or more such chains folded together



Sources of Proteins

found in both animal and plant foods

milk, eggs, seafood, meat, such as chicken and lean beef

soya beans, nuts, grains and vegetables such as French beans

Functions of Proteins

Refer to: [Functions of Biological Molecules](#)

Quick Reference:

- Synthesis of new cytoplasm for growth and repair of worn out body cells
 - Synthesizes enzymes and some hormones
 - formation of antibodies to combat diseases
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Test for Proteins

is done using biurets test

refer to → [Food Tests](#)